Ex.NO:10 Blocks World

AIM:

To transform an initial configuration of stacked blocks on a table into a specific goal configuration.

PROGRAM:

```
initial_state = [ ['A','B'],['C'],[]]
goal_state = [['C','A'],['B'],[]]

def move(state,from_stack,to_stack):
    state[to_stack].append(state[from_stack].po
    return state

def display_state(state):
    for i,stack in enumerate(state):
        print(f"stack {i+1}: {stack}")
    print()

print("Initial State:")
display_state(initial_state)

state = move(initial_state,1,2)
state = move(state,0,1)
state = move(state,0,1)
state = move(state,1,0)
state = move(state,1,0)

print("Final State:")
display_state(state)
```

O/P

Initial State :

Stack 1 : ['A', 'B']

Stack 2 : ['C']

Stack 3 : []

Final State :

Stack 1 : ['A', 'C']

Stack 2 : ['B']

Stack 3 : []

